

Rotary laser thinning of silicon wafer via diffractive optical element: Effects of process parameters on surface integrity

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ABSTRACT

As the demand for computing power and processing speed continues to increase, integrated circuits require greater circuit layer density and better heat dissipation capabilities. Therefore, wafer thinning technology is crucial. Traditional laser thinning technology can obtain a silicon wafer surface with a surface roughness of 749 nm. Here, we show a novel rotary laser thinning technique for silicon wafers utilizing diffractive optical element (DOE) is presented, wherein conventional circular laser spots with Gaussian-distributed energy are transformed into linear spots with uniform energy distribution. This method overcomes critical limitations of conventional laser thinning, offering controllable thinning operations. Experiments demonstrate that DOE-based laser thinning reduces the surface roughness to 196.6 nm with a maximum reduction of 50.34 % amorphous silicon formation and 5.96 % oxide formation. When applied to flat surfaces, DOE-based linear spots effectively mitigate the formation of raised redeposited material and sunken removed materials created by Gaussian spot thinning, resulting in a flat and uniform surface. When applied to uneven surfaces, the method mitigates approximately 50 % of the regenerative effects common in conventional laser thinning. Notably, the surfaces thinned by the DOE-based linear spots exhibit consistent hydrophilicity in both feed and tangential directions, in contrast to the anisotropic wettability of Gaussian-thinned surfaces. The influence of pulse frequency and rotational speed on machining depth and surface roughness was further explored, and optimal thinning parameters were identified. This innovative approach provides a promising alternative thinning method for silicon wafers, with potential applications in semiconductor device fabrication and other high-precision fields requiring smooth and precise surfaces. Furthermore, the multi-pass capability of the DOE-based method shows promising advantages in surface quality control and machining uniformity.

1. Introduction

Silicon (Si) wafers are ubiquitous in the fabrication of integrated circuits within the semiconductor industry. Surface thinning of silicon wafers is of paramount importance, as it is crucial for producing high-quality wafers. The quality of the wafer surface directly impacts the performance and reliability of semiconductor devices. Conventional methods for thinning silicon wafers, such as (i) mechanical thinning/polishing and (ii) chemical mechanical thinning (CMT), as well as

advanced techniques like (iii) electrochemical mechanical thinning (ECMT) and (iv) laser thinning, offer distinct advantages and applications throughout the semiconductor manufacturing process.

For (i), mechanical thinning is one of the earliest techniques used in wafer thinning. It involves the physical removal of material from the wafer surface using abrasive particles and a thinning machine. Despite its straightforward and effective nature, this method may result in surface defects, such as scratches and residual stress, due to the disparity in hardness between the abrasives and silicon [1]. These defects can have

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significant implications for high-precision applications that utilize silicon wafers [2].

For (ii), the CMT process creates a surface with reduced damage and low roughness; however, achieving efficient thinning at room temperature proves challenging due to the chemical properties of silicon, compared to mechanical thinning [3]. Gao et al. [4] achieved a wafer surface roughness of 0.259 nm by enhancing the abrasive structure. Zhao and Lu [5] successfully reduced the surface roughness to 0.05 nm by optimising the silicon slurry and thinning process. Nevertheless, efficiency remains a primary concern, as the material removal rate only reached a few microns per hour [3], despite attempts made in modifying processing conditions [6], catalysts addition [7], and chemical solution modification [8].

For (iii), to achieve efficient thinning, ECMT was proposed as a method that uses the thinning material as the anode for the oxidation reaction, subsequently eliminating the oxide layer [9]. Under specific solutions and conditions, ECMT demonstrated a removal rate exceeding 10 $\mu\text{m}/\text{h}$, which was dozens of times faster than that of CMT [10]. However, ECMT has struggled to gain widespread use due to challenging processing conditions and limitations in processing size [11].

For (iv), laser thinning has emerged as a state-of-the-art technology for silicon wafer thinning. The precise control and uniform energy distribution of the laser beam enable accurate material removal with minimal damage, making it particularly advantageous for applications that require smooth surfaces and minimal damage, such as in the fabrication of high-performance photonic devices and semiconductor components [44]. Keiichiro et al. [12] achieved flat surfaces with a surface roughness close to 1 nm by increasing the pulse width and decreasing the peak laser irradiance. Subsequently, nanosecond pulsed lasers were used as post-processing tools for mechanical thinning, reducing surface roughness from 12 nm to 8 nm [13]. Huang et al. [11] reduced silicon wafer roughness to 0.08 nm using nanosecond laser annealing technology. Despite these efforts, most studies have focused on lasers with Gaussian-distributed energy, which has led to several challenges in achieving high-quality laser machining (see Section 2.1). In laser thinning, a Gaussian distribution means that the intensity of the laser beam is highest at the centre and gradually decreases towards the edges.

To address the aforementioned issues, this paper introduces a novel rotating laser thinning method based on diffractive optical elements (DOEs). By utilizing DOE as an advanced technology for beam shaping, it enables the creation of customised beam shapes, uniform energy distribution, and adjustable focal positions, offering precise control over the phase and amplitude of light. The integration of DOE significantly enhances the flexibility and efficiency of laser thinning processes, thereby overcoming the drawbacks associated with Gaussian lasers and meeting the demands for high precision and complexity in advanced manufacturing.

The paper begins by presenting a new DOE component and a customised platform (see Sections 2.3 and 3). A comparative analysis is conducted between flat silicon wafer surfaces thinned using DOE-based and Gaussian (GAU) lasers, considering surface morphology (see Section 4.1), roughness (see Section 4.2), material behaviours (see Section 4.3), and surface wettability (see Section 4.4). Subsequently, the distinct ablation mechanisms of DOE-based techniques, in comparison with GAU laser beams, are elucidated in Section 4.5. Furthermore, non-flat surfaces are individually thinned using DOE-based and GAU lasers, followed by a comparative study presented in Section 5. Section 6 investigates the optimal processing parameters. Brief conclusions, along with the results achieved in this paper, are drawn in Section 7.

2. Rotary laser thinning based on diffractive optical elements

2.1. Challenges in conventional linear & rotary laser thinning

Conventional laser thinning [14] typically uses a galvanometer

system for precise control of laser focus movement within the focal plane, enabling line-by-line thinning of the silicon wafer surface (see Fig. 1a). A zigzag scanning path is commonly employed (see Fig. 1b). However, this approach may result in two surface issues (see Fig. 1c):

- (i) The formation of pits is observed at the turning points of the zigzag scanning path (see Fig. 1d) due to the acceleration and deceleration of the galvanometer motor at the start and end of each trajectory. This results in overlapping laser pulses ablation a localized area, leading to overheating effects. The ablation grooves formed at these turning points are deeper than those along the straight paths, thereby degrading the quality of the thinned surface.
- (ii) Visible trenches are generated between adjacent scanning paths due to the Gaussian energy distribution at the focal spots. This energy distribution causes the laser to ablate the central region with a greater removal depth and the edges with a lesser removal depth, resulting in discernible trenches that also affect the surface quality of the machined material (see Fig. 1e).

To address these limitations, Tan et al. [15] propose an improved laser thinning method known as conventional rotary laser thinning (CRLT). In this method, the silicon wafer is placed on a uniformly rotating turntable, while the laser beam moves linearly from the outer periphery towards the centre of rotation (Fig. 1f). This approach results in an arc-shaped thinning path, effectively mitigating pit formation caused by the acceleration and deceleration of the pulsed trajectory (see Fig. 1g, h). Despite this improvement, trenches still emerge between adjacent scanning paths (see Fig. 1i). This can be attributed to non-uniform ablation pits formed due to the Gaussian energy distribution within a single pulse footprint. The shape of the removed material at the pit centre is semi-circular in the cross section, whereas at the pit edges, there are trenches of redeposited material, creating a regular wavy drop-off on the wafer surface [16]. These defects would (i) affect the smoothness and roughness required for high-precision manufacturing processes, potentially having a negative impact on subsequent thinning processes, and (ii) the redeposited material may have a different composition from the original wafer, introducing impurities and potentially affecting the electrical and optical properties of the device.

2.2. Rotary laser thinning based on diffractive optical elements

To address the aforementioned issues, we propose a novel laser thinning technique that employs a DOE to transform the laser beam into an elongated spot. This transformed spot is then focused onto the wafer surface through a focusing lens.

The wafer is fixed on a rotating platform that rotates with a constant angular velocity, ω . To ensure that the laser spots are spaced exactly D apart in the direction of rotation without overlapping, the initial rotational speed, ω , can be determined based on this condition. The number of laser pulses, N , at the outermost edge of the disk during one full rotation can be calculated using Eq. (1).

$$N = f \cdot T \quad (1)$$

where f is the laser pulse frequency, and T is the time required for the platform to complete one rotation. The time T can be related to the angular velocity ω through Eq. (2)

$$T = \frac{2\pi}{\omega} \quad (2)$$

Using the disk circumference and the distance D between adjacent laser spots, the number of pulses N at the outermost edge can be calculated as shown in Eq. (3)

$$N = \frac{2\pi r}{D} \quad (3)$$

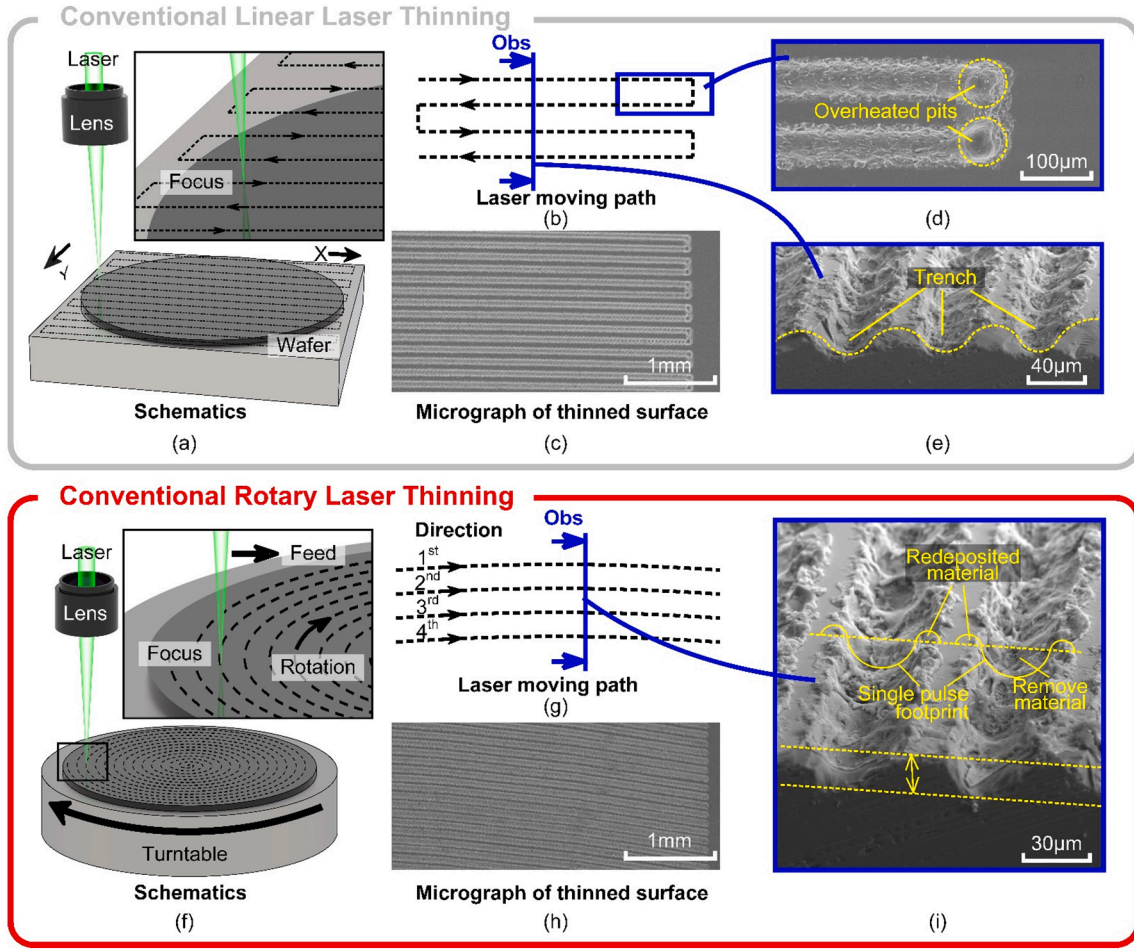


Fig. 1. Conventional linear laser thinning (a) Schematic diagram of a wafer thinned using a galvanometer-moving laser spot; (b) Trajectory of the laser spot movement achieved through a galvanometer-moving laser; (c) Visible trajectory topography of a galvanometer-thinned wafer using scanning electron microscope observation (SEM); (d) Overheated crater topography formed at an inflection point; (e) Highly protruding topography between neighbouring scanning paths; Conventional rotary laser thinning (f) Schematic diagram of a wafer thinned using a turntable; (g) Spot movement trajectory for rotary laser thinning; (h) Using scanning electron microscopy to analyse the visible trajectory topography of a wafer thinned by rotary laser thinning (SEM); (i) Trenches still formed between neighbouring scanning paths.

where r is the radius of the disk, and D is the width of the laser spot, which is also the distance between adjacent pulses. Consequently, the rotational speed ω can be determined from Eq. (4)

$$\omega = f \cdot \frac{D}{r} \quad (4)$$

In the feed direction, to ensure that each laser spot also touches exactly in the feed direction, the feed speed v must be dynamically adjusted based on the processing time. The laser moves a distance L in the feed direction during one full rotation. Therefore, the initial feed speed v is given by Eq. (5)

$$v = \frac{L}{T} = \frac{L\omega}{2\pi} \quad (5)$$

However, as the platform radius r decreases, the linear velocity of the spots in the rotation direction also decreases. To maintain precise contact between spots in the feed direction, the feed speed must be adjusted in real-time as the process progresses. Thus, the real-time feed speed v_f is given by Eq. (6)

$$v_f = v \frac{r}{r - L\omega t} \quad (6)$$

where L is spot length and t is processing time. Therefore, the real-time feed speed v_f increases as the processing time increases, ensuring that

the laser spots remain precisely aligned in the feed direction.

This ensures uniform scanning of the entire wafer surface, as depicted in Fig. 2a. Compared to achieving a Gaussian spot without employing the DOE, our method maintains the same spot width, D , but significantly increases its length, L , by a factor of 10. Consequently, with identical processing parameters and duration, our linear spot covers a considerably larger area than a Gaussian spot, greatly enhancing thinning efficiency (see Fig. 2b).

The laser beam emitted from the source initially passes through a beam expander, which enlarges its diameter to 8 mm, effectively reducing beam divergence and producing a high-quality beam that closely resembles an ideal parallel beam (see Fig. 2c). At this stage, zone 1 of the energy distribution is concentrated at the centre, with intensity decreasing towards the periphery. The beam then undergoes three mirror reflections, redirecting it perpendicular to the thinning plane before passing through the DOE and focusing lens. As the beam gradually converges, zone 2 represents the energy distribution after passing through the DOE, causing the spot shape to gradually transform from circular to linear while achieving a progressively uniform energy distribution. Finally, upon focusing on the surface of the workpiece, a linear spot is formed, with its corresponding energy distribution depicted in zone 3 (see Fig. 2d).

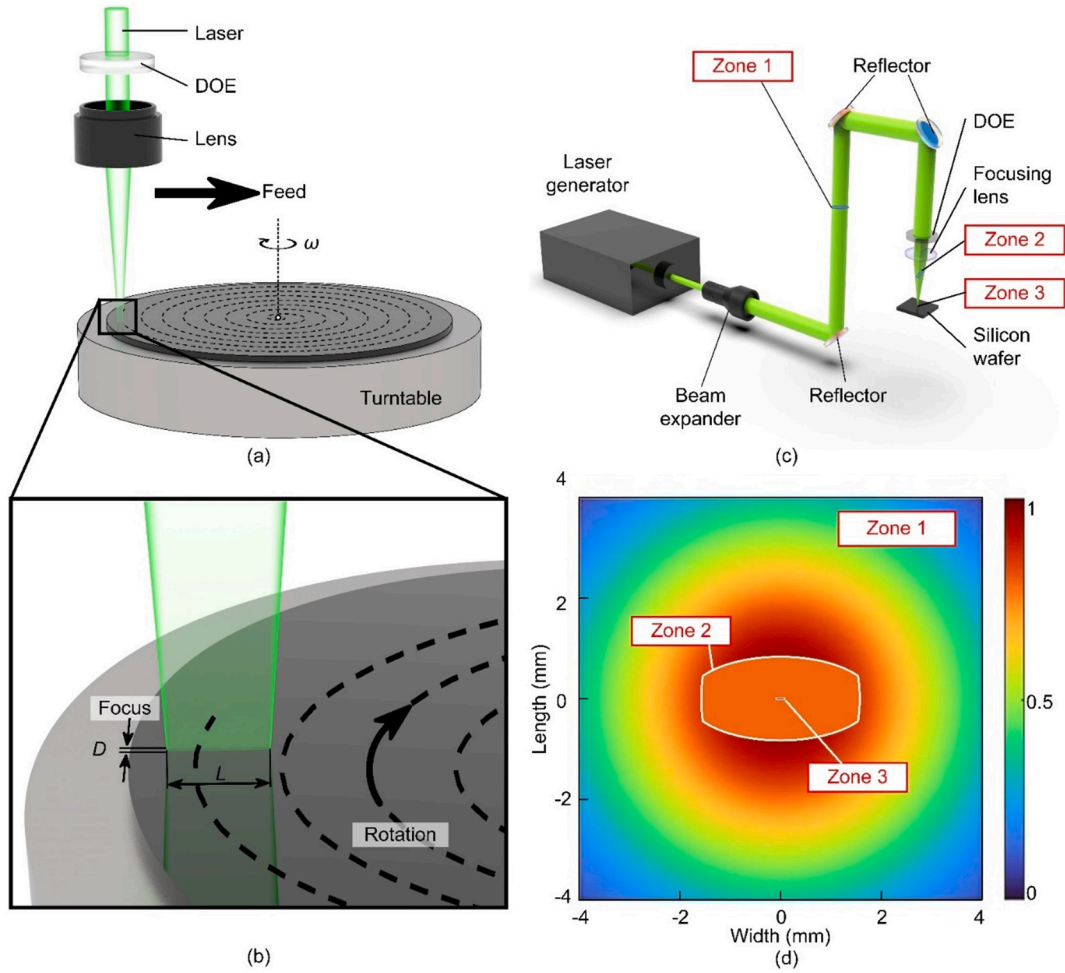


Fig. 2. (a) Schematic diagram of DOE-based laser thinning; (b) Spot shape on the wafer surface; (c) DOE-based laser thinning optical path diagram; (d) Spot size and beam energy distribution at different positions.

2.3. DOE design

To meet the requirements for laser spot shape and energy distribution, a DOE is designed to produce linear spots (see Fig. 3a). By precisely etching microstructures on its surface, a functional diffraction grating is formed (see Fig. 3b). These finely engineered gratings induce phase shifts in the incident light as it passes through the DOE, resulting in varying optical path lengths at different spatial locations within the beam, which in turn leads to distinct phase delays. Interference occurs during the propagation of the subsequent modulated beam, causing a redistribution of energy in the spatial domain. Finally, after passing through a focusing lens, a uniform linear spot forms at the target plane (see Fig. 3c).

Designing optical diffraction elements typically offers multiple possible solutions. Although the input light parameters (beam diameter, wavelength) and target spot characteristics (shape, energy distribution) are fixed, the design of a diffraction grating involves several degrees of freedom, such as the grating period, depth, shape, and material. These parameters can be adjusted and optimised, leading to different grating structures that meet the same design requirements. Therefore, the design of the DOE structure is not necessarily unique.

The specific control methods can be referenced in Eqs. (7)–(10). The relationship between the DOE grating structure and energy distribution can be quantitatively controlled through optical design. Parameters such as the DOE grating period, depth, and incident angle all influence diffraction efficiency and energy distribution. By employing diffraction theory and numerical simulations, the energy distribution for different

grating structures can be predicted. In the specific DOE design presented in this paper, a grating structure with a sawtooth shape is utilized, comprising an array of flat and sawtooth-like grooves etched on the surface (see Fig. 3d).

- (i) The groove spacing plays a pivotal role in determining the diffraction angle of diffracted beams, as described by the grating equation (see Eq. (7))

$$d \sin \theta = n \lambda \quad (7)$$

where d is the distance between adjacent engraved lines, θ is the diffraction angle, $\lambda = 532 \text{ nm}$ is the wavelength of the light wave, and $n = 1$ is the diffraction order. By carefully selecting the groove spacing and profile, we can precisely control both the diffraction angles and intensities of the resulting beams, directing the diffracted light to constructively interfere and form the desired linear beam profile with a uniform energy distribution (see Fig. 3e). The diffraction angle θ can be calculated by Eq. (8)

$$\theta = \tan^{-1} \frac{r - a}{2f} = \tan^{-1} \frac{8 - 0.02}{2 \times 75} = 0.673^\circ \quad (8)$$

where $r = 8 \text{ mm}$ is the diameter of the incident light, $a = 20 \mu\text{m}$ is the width of the spot target shape, and $f = 75 \text{ mm}$ is the focal length of the focusing mirror.

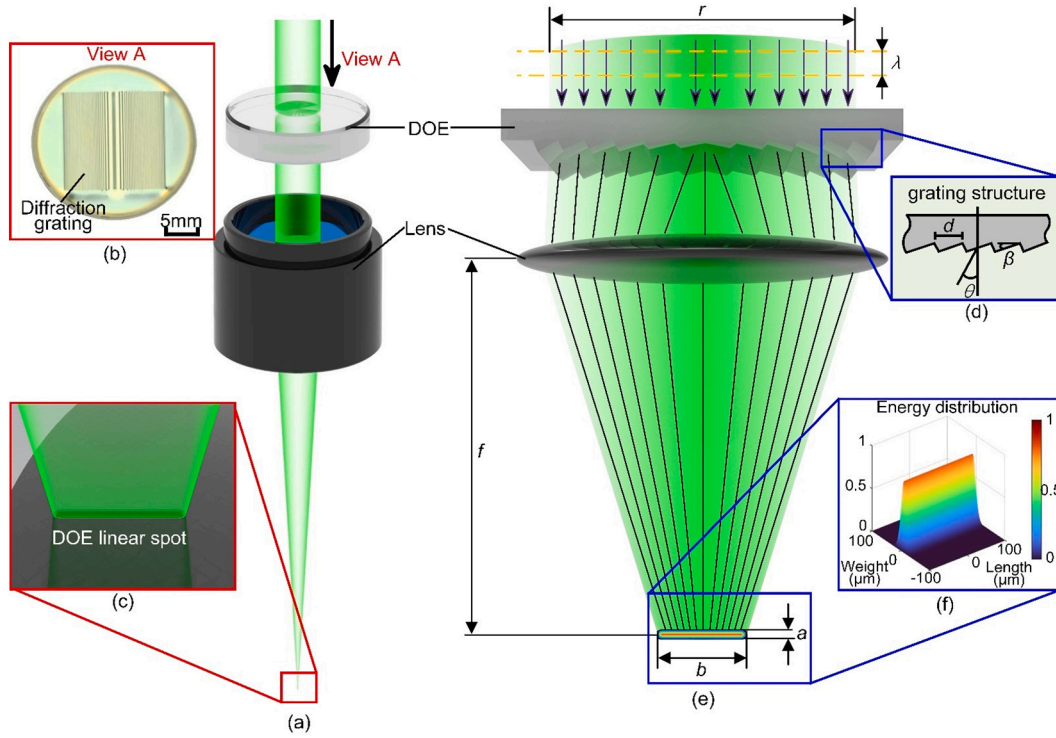


Fig. 3. (a) Physical figure of DOE; (b) Shape of DOE under orthogonally polarized light; (c) Shape of the focused spot after the laser passes through the DOE; (d) Microstructure of DOE surface grating; (e) Diffraction schematic diagram of the laser passing through the DOE; (f) Energy distribution of the focused spot.

- (ii) The groove angle plays a crucial role in controlling the angular distribution of the diffracted light, influencing the overall shape and uniformity of the resulting beam profile. As light interacts with the slanted facets of the grooves within the DOE, it undergoes refraction according to Snell's Law [17]. In this context, Snell's Law establishes a relationship between the angles of incidence and refraction based on the refractive indices of both media involved. For our DOE design, we express Snell's Law as Eq. (9)

$$m \sin \beta = \sin(\theta + \beta) \quad (9)$$

where $m = 1.76$ is the refractive index of sapphire, and β is the inclination angle of the inclined surface of the diffraction grating groove. Therefore, the inclination angle of the inclined surface of the diffraction grating groove β can be calculated by Eq. (10)

$$\beta = \tan^{-1} \frac{\frac{n_d^2}{d}}{m - \sqrt{1 - \left(\frac{n_d^2}{d}\right)^2}} \approx 0.42^\circ \quad (10)$$

This relationship demonstrates that the collective interaction of multiple gratings results in the generation of a composite diffraction pattern, which arises from the predetermined diffraction behaviour exhibited by each individual grating. By manipulating the shape and spacing of microscopic gratings on the DOE, we can precisely control both the intensity and direction of diffraction, thereby achieving the desired linear light output. Consequently, a linear spot with a consistent energy distribution along its length and a Gaussian distribution in its width is obtained (see Fig. 3e). This spot shape ensures uniform energy in the feed direction and, in conjunction with rotational laser thinning, can effectively remove trenches between neighbouring machining paths, resulting in a flatter surface with lower roughness. Based on the above design approach, the energy distribution width can be controlled using a mathematical model before fabrication. Once manufactured, the resulting beam spot is well-defined and predictable.

2.4. DOE manufacturing process

In this study, we use a sapphire substrate as the fabrication platform for the DOE, employing various advanced microfabrication techniques to precisely create the wedge-shaped grating structure (see Fig. 4). The detailed fabrication process and the purpose of each step are as follows:

- (i) **Substrate Preparation** (see Fig. 4 i): The sapphire substrate is first prepared as the platform for the DOE, ensuring it is clean and free from contaminants before further processing.
- (ii) **Photoresist Coating** (see Fig. 4 ii): A positive photoresist (AZ 5214E) is uniformly coated onto the sapphire substrate using the spin-coating method. The photoresist thickness is $5 \mu\text{m}$, with optimised spin speed and time to ensure a uniform and smooth coating, providing a solid foundation for the subsequent pattern exposure.
- (iii) **Soft Bake Process** (see Fig. 4 iii): The substrate is then subjected to a soft bake process at 100°C for 2 min to remove the solvent from the photoresist, ensuring its stability and strong adhesion to the substrate, thereby providing a stable photoresist matrix for the subsequent exposure process.
- (iv) **Pattern Exposure via Laser Direct-Write** (see Fig. 4 iv): A laser direct-write technique is used for pattern exposure. This technique precisely controls the position and intensity of the laser beam to etch the designed grating pattern. For the wedge-shaped grating, the laser direct-write enables precise adjustments of the depth and period of the photoresist layer, ensuring the complexity and accuracy of the grating structure. In this process, a laser with a $1 \mu\text{m}$ spot size, 405 nm wavelength, and 20 mW power is used for exposure, typically for several minutes to ensure the required exposure dose.
- (v) **Development Process** (see Fig. 4 v): After exposure, the sample undergoes a development process. The purpose of development is to remove the exposed photoresist through a chemical reaction, leaving the unexposed regions intact, thereby forming the grating

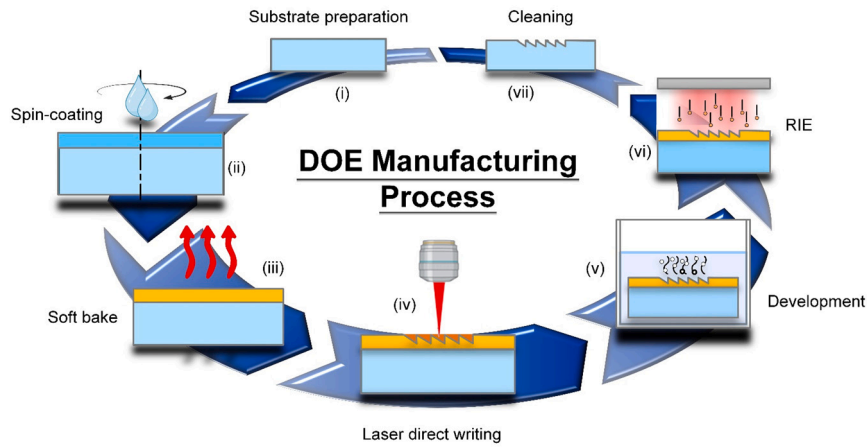


Fig. 4. DOE manufacturing process.

mask pattern. For positive photoresist, the exposed areas of the photoresist dissolve, creating the desired grating shape. The development time is 3 min, followed by rinsing with deionised water.

- (vi) Reactive Ion Etching (RIE) (see Fig. 4 vi): The sample is then subjected to reactive ion etching (RIE). During this step, RIE technology simultaneously etches both the photoresist layer and the sapphire substrate. Specifically, ion bombardment and chemical reactions remove the exposed photoresist, while the regions of the sapphire substrate not protected by the photoresist

are etched away. As a result, the grating structure is transferred onto the sapphire substrate, ensuring the formation of the precise wedge-shaped grating structure. Chlorine ions (Cl^-) are used as etching ions, and the etching time is adjusted according to the desired depth.

- (vii) Cleaning Process (see Fig. 4 vii): After etching, a cleaning process is performed. The purpose of this step is to remove any residual photoresist and contaminants, ensuring the substrate surface is clean and free from impurities, ultimately resulting in the final diffractive optical element.

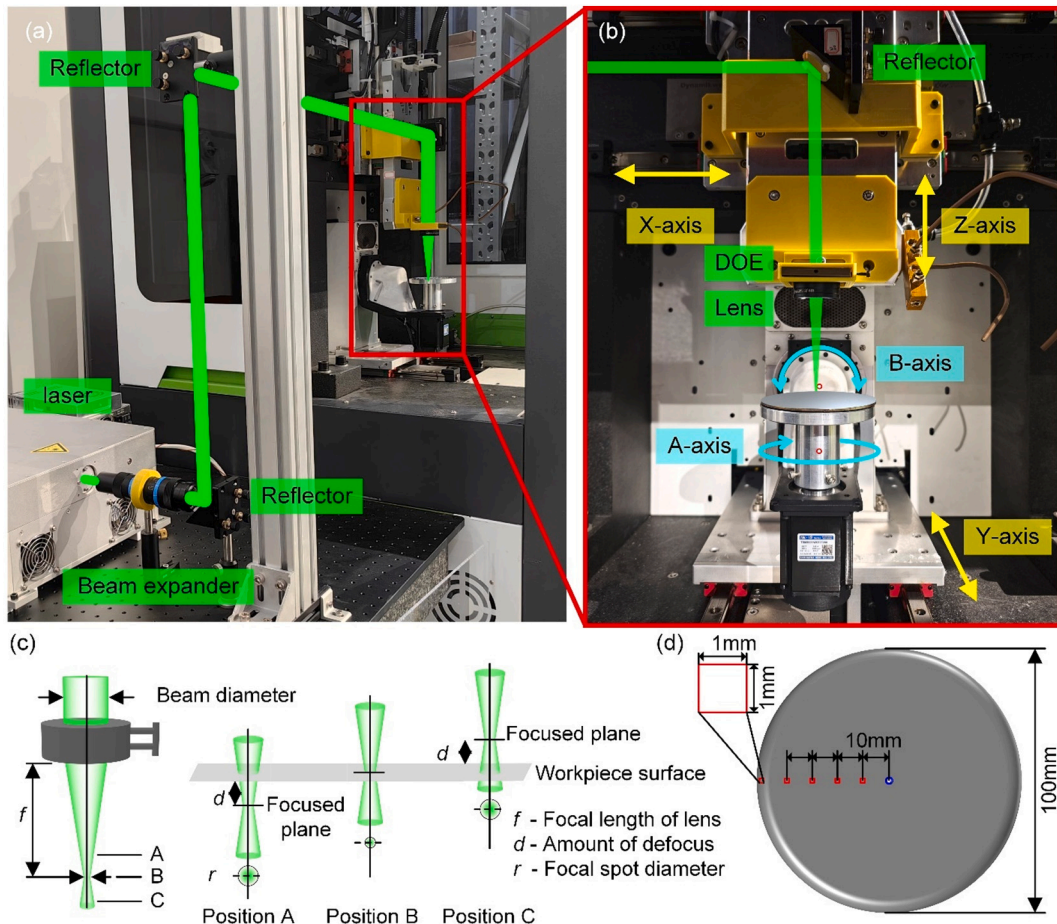


Fig. 5. (a) Laser thinning platform; (b) Platform details and laser head; (c) Effect of laser focus defocus.

3. Experiment platform

A five-axis laser thinning platform incorporating DOE technology has been developed (see Fig. 5a and b). The laser beam is expanded by a beam expander and guided through a series of three reflective mirrors before reaching the DOE and focusing lens assembly. The X-axis of the machine tool carries the Z-axis, enabling both horizontal and vertical movement of the focusing lens. Meanwhile, the Y-axis is equipped with both A and B axes, facilitating forward/backward translation and rotation of the workpiece. This integrated platform provides sufficient degrees of freedom and precise control to effectively utilize the line-shaped beam generated by the DOE for high-quality and efficient laser thinning applications.

To conduct laser experiments in an atmospheric environment using this platform, a standard single-sided thinned sample with dimensions of $10 \times 10 \times 0.6$ mm was selected. The sample was irradiated using a picosecond YAG laser system operating at a wavelength of 532 nm and a pulse width of 50 ps, with the number of pulses regulated by a computer-connected pulse generator.

The experimental setup ensures precise focal spot alignment of the laser beam with the material surface. This approach is preferred over positive defocus (where the focal spot is above the surface) due to its adverse effects on energy density, which results in shallower cuts and potentially increased surface roughness caused by energy dispersion. Similarly, negative defocus (where the focal spot is below the surface) is avoided as it reduces thinning efficiency and expands the heat-affected zone, despite a temporary increase in energy density that may lead to material deformation and micro-cracks. Therefore, maintaining the focal spot on the material surface optimises both energy density and thinning outcomes, ensuring efficient and high-quality material removal (see Fig. 5c).

This study compares surface morphology, surface roughness, material behaviour, residual stress, and surface hydrophilicity/hydrophobicity using GAU and DOE spot patterns. The process parameters utilized are presented in Table 1. Additionally, an optimisation of the process parameters for DOE is conducted, with the employed parameters illustrated in Table 2.

In this paper, a range of state-of-the-art instruments was employed to comprehensively analyse the effects of the laser thinning process. A scanning electron microscope (SEM, EVO 18, ZEISS, Germany) was utilized to observe the surface morphology of the samples, identifying the microstructural changes induced by the laser treatment, and was combined with an energy dispersive spectrometer (EDS, EVO 18, ZEISS, Germany) to gather information on the elemental composition. An X-ray diffractometer (XRD, D8 Advance, Bruker Corp., Germany) was used to analyse the crystal structure, assess phase changes, and evaluate structural stability. Additionally, an ultra-depth-of-field microscope (Model VHX-6000, Keyence Corp., Japan) provided high-resolution three-dimensional surface images, enabling detailed observation of surface features such as roughness and defects. The wettability of the samples was assessed using a contact angle meter to determine the impact of laser treatment on surface hydrophilicity. Finally, an atomic force microscope (AFM, FastScan Bio, Bruker Dimension, Germany) was employed to obtain nanoscale information on surface morphology, accurately measure surface roughness, and observe microstructural changes. To ensure the accuracy of the experimental data, a 1×1 mm area was measured at five different locations, each spaced 10 mm apart

Table 1
The experimental parameters of GAU and DOE compared.

Group	1	2	3
Spot type	DOE, GAU	DOE, GAU	DOE, GAU
Single pulse energy (μ J)	34,37,43,48,53,58	53	53
Process time	1	1,5,10,50,100,200	1
Rotational speed (rpm)	229	229	115,229

Table 2

DOE process parameters.

Group	1	2
Spot type	DOE	DOE
Single pulse energy (μ J)	37,43,48,53	43
Rotational speed (rpm)	57	57,115,172,229

from the previous one. Measurements were taken three times at each location, and the average value was calculated (see Fig. 5d). The integration of these instruments facilitated a thorough analysis of the laser thinning process and provided comprehensive data support.

4. Flat surface thinning with DOE-based laser compared to GAU laser

This section compares the thinned flat surfaces using a DOE-based laser beam and a GAU laser beam from four aspects: (i) machined surface morphology (Section 4.1), (ii) surface roughness (Section 4.2), (iii) material behaviours (amortization and oxidation) (Section 4.3), (iv) surface wettability (Section 4.4), and (v) the ablation mechanisms of the DOE-based technique compared with the GAU laser beam (Section 4.5).

4.1. Machined surface morphology

The flat surface thinned by the GAU laser beam exhibited distinct grooves with bowl-shaped cross-sectional profiles, where the centre of each individual groove was deeper than both sides. The combination of these bowl-shaped grooves resulted in a prominent surface wave pattern, characterised by a uniform and regular arrangement of peaks and valleys. These phenomena were consistently observed across various parameters (see Fig. 6a, c, e, and g). This can be attributed to the inherent Gaussian intensity profile of the GAU laser beam and the overlapping ablation pits. The central region of the laser beam possessed the highest energy intensity, causing deeper material ablation and leading to the formation of pronounced notches at its deepest point. As the beam energy intensity decreased towards the edges, the depth of ablation also diminished, resulting in the bevelled edges characteristic of Gaussian profiles. Consequently, as successive grooves overlapped during laser scanning on the material's surface, they generated a consistent undulating pattern.

In contrast to the GAU beam, the grooves produced using the DOE exhibited a flat-bottomed interface with a consistent ablation depth across each groove. These flat-bottomed grooves maintained uniform surface profiles regardless of the varied parameters employed (see Fig. 6b, d, f, and h). This is because the uniform energy distribution of the DOE-based linear laser spot over the entire cross-section results in consistent material ablation with no peaks or valleys. This uniformity minimises the effect of overlapping channels, leading to accurate and consistent material removal. The uniform energy distribution provided by the DOE-based beam reduces the risk of collateral damage, ensures cleaner ablation, and diminishes the need for post-processing steps.

Compared to the GAU spot, the surface processed with the DOE spot demonstrates a significant advantage in flatness under the same processing parameters. Cross-sectional analysis of the surfaces reveals that the DOE-processed surface is not only flatter but also has a markedly lower roughness than the GAU-processed surface, with flatness improvements exceeding seven times. The surface processed with the GAU spot exhibits pronounced peaks and valleys due to the high-energy centre of the beam, whereas the uniform energy distribution of the DOE spot eliminates such irregular undulations, ensuring a more even and consistent surface after processing.

4.2. Surface roughness

In the semiconductor industry, high product yields require careful

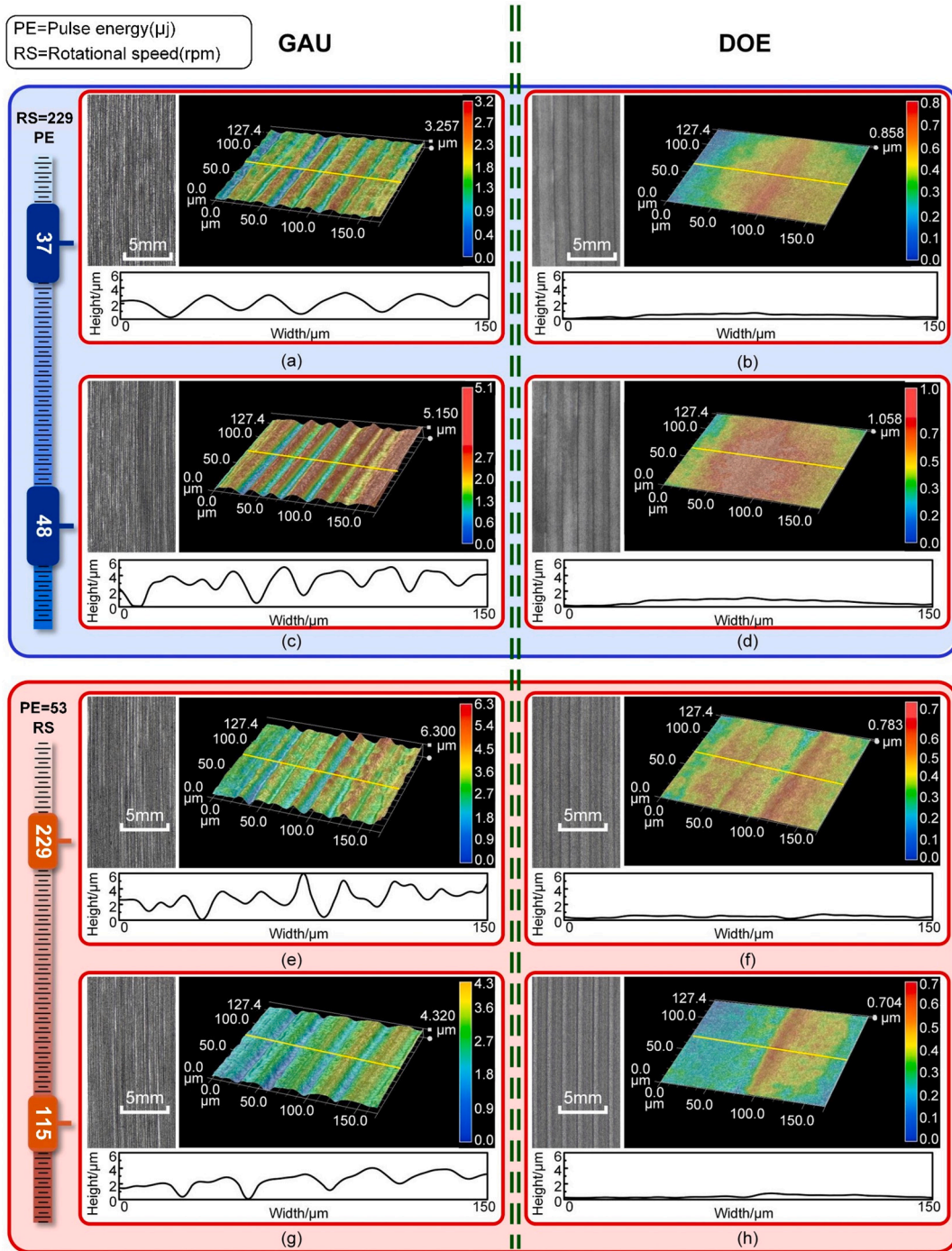


Fig. 6. When the pulse energy is 37 μJ and the rotational speed is 229 rpm, (a) The surface morphology resulting from treatment with GAU laser beam is presented; (b) Surface morphology observed on the silicon wafer after DOE-based laser treatment is illustrated. At a pulse energy of 48 μJ and a rotational speed is 229 rpm, (c) The wafer's surface morphology following GAU laser treatment is shown; (d) Characteristics displayed on the silicon wafer after being treated with DOE-based laser beam are analysed. When the pulse energy was 53 μJ and the rotational speed is 229 rpm, (e) The morphology of the wafer after treatment with a GAU laser beam is presented; (f) Surface features observed on the silicon wafer treated with DOE-based laser beam are illustrated. At a pulse energy of 58 μJ and a rotational speed is 115 rpm; (g) The surface morphology resulting from GAU laser treatment is shown; (h) Characteristics of the silicon wafer thinned using DOE-based laser beam are analysed.

attention to even the slightest variations in surface roughness, as these can significantly affect device performance. The maximum surface roughness (R_z) is chosen for its heightened sensitivity to variations in surface peaks and valleys, making it a more precise indicator of the microstructure requirements for semiconductor products compared to other roughness parameters. Notably, regardless of changes in laser parameters, the DOE-based beam consistently yields lower values for

surface roughness (R_z) compared to those obtained with the GAU beam:

(i) As the single-pulse energy increases, the DOE-based beam induces a significantly smaller increase in machined surface roughness compared to the GAU beam. At 34 μJ , there was a difference of approximately 50 nm in surface roughness between the two methods. However, when the single-pulse energy increased to 58 μJ , this difference grew to around 450 nm (see Fig. 7a). This can be attributed to the

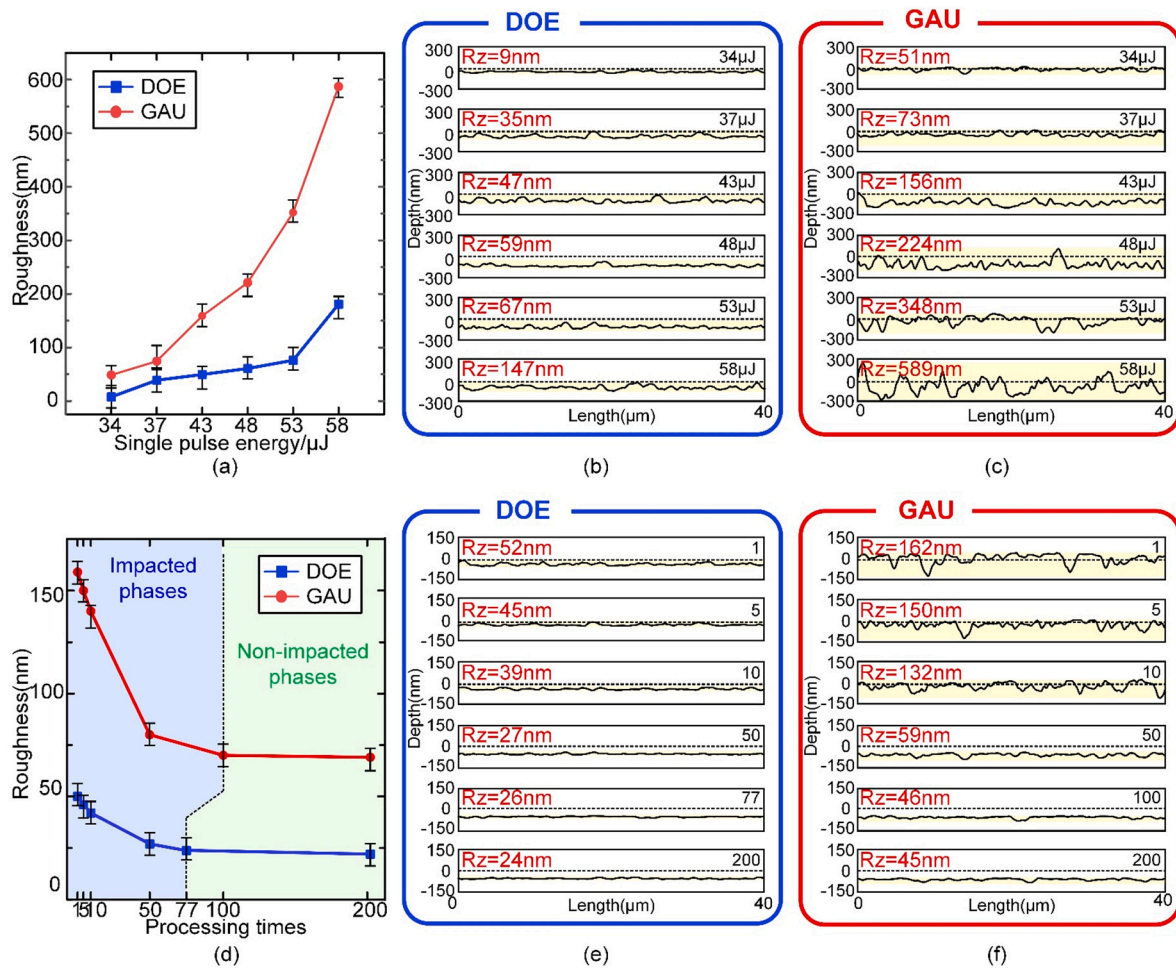


Fig. 7. (a) The impact of varying single pulse energies on roughness; (b) Surface morphology observed in silicon wafers thinned by DOE-based laser beam at different single pulse energies; (c) Examination of surface morphology in silicon wafers thinned using GAU laser beam with varying single pulse energies; (d) The influence of different thinning times on roughness; (e) Morphologies seen in silicon wafers thinned with DOE-based laser beam at various thinning times; (f) Analysis of the morphology in silicon wafers thinned by GAU laser beam across different thinning durations.

fact that the primary factor affecting surface roughness is the sputtering of redeposited material during thinning. Specifically, the uniform energy distribution of the DOE-based laser beams reduces localized high-energy hotspots and non-uniform removal and redeposition of material, leading to a smaller increase in surface roughness. In contrast, the concentrated energy peaks of the GAU laser beams result in high energy densities on the material surface, causing rapid vaporization and ejection of material, which leads to deeper craters and pronounced surface irregularities, thereby increasing surface roughness (see Fig. 7b, c).

(ii) As the number of thinning passes increases, the DOE-based beam exhibits a significantly less pronounced reduction in surface roughness compared to the GAU beam. After one pass, there was approximately a 120 nm difference in surface roughness between the two methods. However, when the number of thinning passes increased to 50, this difference decreased to about 50 nm (see Fig. 7d). This phenomenon can be explained by the absence of pronounced peaks and valleys in the DOE-based beam's energy profile, which minimises the formation of deep craters and facilitates smoother surface evolution with each subsequent pass. Therefore, the uniform energy distribution provided by the DOE-based beam ensures a consistent and gradual material removal process with every pass (see Fig. 7e). On the other hand, multiple passes are required for effective ablation and smoothing due to the deep craters and sharp features generated by the GAU beam (see Fig. 7f).

(iii) The variation in surface roughness of the workpiece can be divided into two phases: the impacted phase and the non-impacted

phase. In the impacted phase, an increase in thinning passes for both the DOE-based and GAU-based beams leads to a decrease in surface roughness of the workpiece. However, in the non-impacted phase, the surface roughness remains unchanged despite further increases in the number of passes. It is observed that DOE-based laser beam thinning reaches the non-impacted phase more quickly compared to GAU laser beam thinning. Specifically, after 77 thinning passes, the DOE-based laser beam enters the non-impacted phase, while the GAU laser beam requires 100 passes (see Fig. 7d). This phenomenon can be attributed to effective ablation occurring within a specific range above and below the laser focus, where the laser energy density exceeds the wafer's ablation threshold. The farther the laser spot is from the focus, the lower the energy density. In multiple thinning processes, each pass removes a certain thickness of material. If the remaining material's surface falls outside the effective ablation range, no further ablation occurs due to insufficient energy density from the laser. The faster transition to the non-impacted phase for the DOE-based beam can be explained by its smaller defocusing and higher maximum energy density compared to the GAU beam.

As a result, the utilization of DOE-based beams ensures minimal surface roughness by allowing a broad range of single-pulse laser energies or by achieving low surface roughness with a reduced number of thinning passes. This provides large tolerances and enhances the efficiency of the laser thinning process.

4.3. Material behaviours (amortization and oxidation)

The formation of amorphous silicon and silicon dioxide (SiO_2) on the surface of the material during laser thinning significantly impacts the electrical and thermal conductivity of the wafer. These effects not only influence the physical properties but also have important implications for subsequent processing and applications. The X-ray diffraction (XRD) analysis of the initial single-crystal silicon surface revealed a sharp and intense diffraction peak at 28.4° (see Fig. 8a Initial Surface). In contrast, the diffraction pattern of the thinned single-crystal silicon surface showed a broad and low-intensity peak, indicating a significant reduction in crystallinity, with a crystallinity index of only 0.68 % (see Fig. 8a, 34 μJ).

DOE-based laser beam thinning reduced the generation of amorphous silicon. Although the surface crystallinity of the DOE laser beam was initially lower than that of the GAU laser beam after a single thinning pass at a single-pulse energy of 34 μJ , it surpassed the crystallinity of the GAU beam as both the single-pulse energy and the number of thinning passes increased. The crystallinity difference reached a maximum of 60.34 % at 43 μJ and 29.38 % after 100 thinning passes, respectively (see Fig. 8a and b). This phenomenon can be attributed to the uniform energy distribution provided by the DOE-based linear spot, which allows for consistent ablation across the entire surface, ensuring that each area reaches the recrystallization threshold for single-crystal silicon. In contrast, the unique GAU energy distribution pattern results

in only the central region of the laser beam reaching the ablation threshold, while the peripheral areas generate a larger amount of amorphous silicon.

DOE laser thinning also significantly reduced the formation of SiO_2 . Regardless of the single-pulse energy and the number of thinning passes, the surface treated with the DOE laser beam consistently exhibited a lower percentage of elemental oxygen compared to the GAU laser beam. Additionally, an increase in both single-pulse energy and the number of thinning passes led to a greater disparity in elemental oxygen mass percentage between the two thinning methods. For instance, at a single-pulse energy of 58 μJ , the elemental oxygen content for the DOE-based beam was 3.01 %, while for the GAU-based beam it was 5.92 %, representing a difference of 2.91 %. After 500 thinning passes, the elemental oxygen content for the DOE-based beam was 3.56 %, whereas for the GAU-based beam it was 9.52 %, resulting in a difference of 5.96 % (see Fig. 8c and d). This can be attributed to the uniform energy distribution of the DOE-based laser beam, which prevents localized overheating, reduces sensitivity to variations in single-pulse energy, and minimises oxidation reactions caused by the intense local heating associated with the Gaussian energy distribution of GAU laser beams.

As a result, DOE-based laser thinning enhances crystallinity and effectively reduces oxidation, preserving the desired electrical, mechanical, and chemical properties of single-crystal silicon.

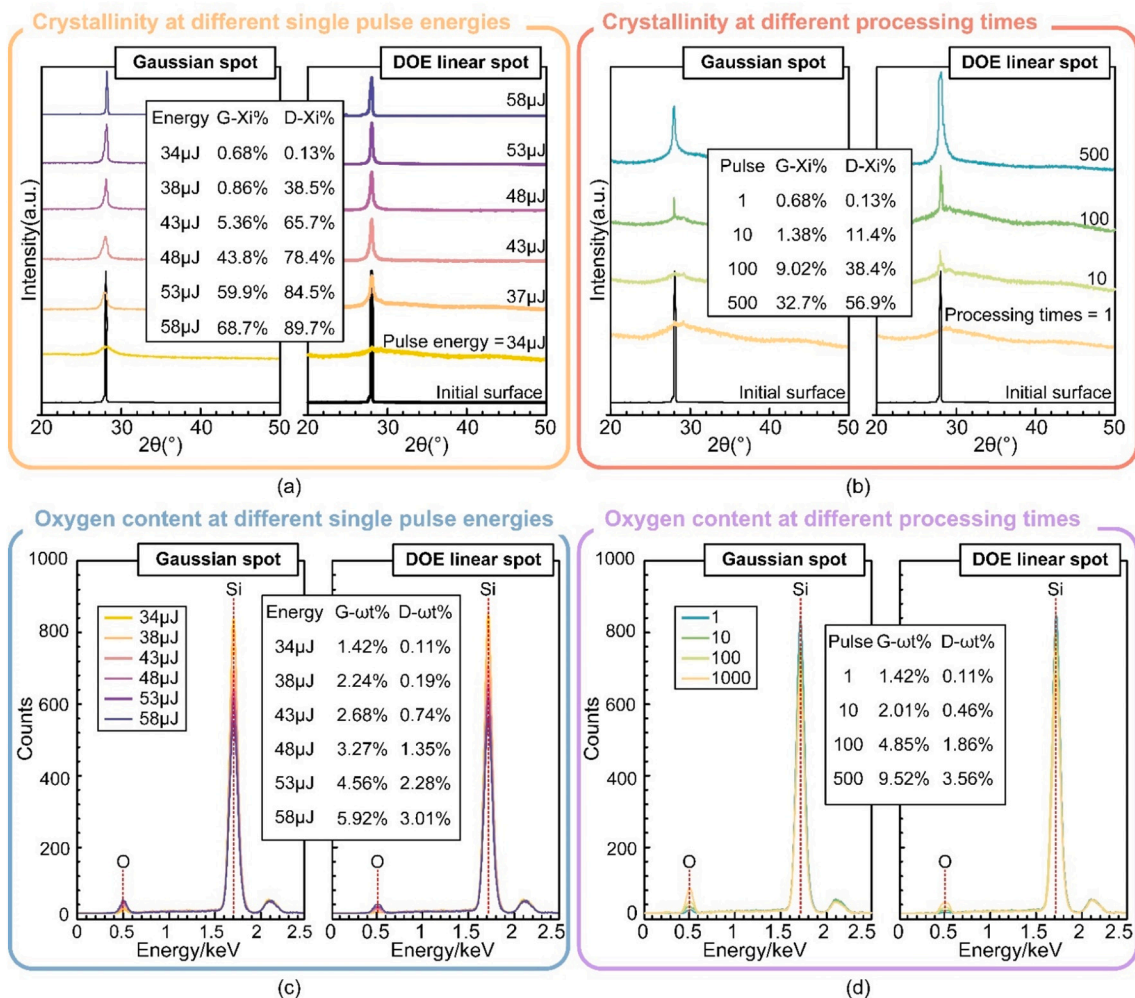


Fig. 8. Comparison of crystallinity between Gaussian spot and DOE-based linear spot, (a) Analysis under varying single pulse energies; (b) Examination across different thinning times; Comparison of oxygen content between Gaussian spot and DOE-based linear spot, (c) Assessment based on different single pulse energies; (d) Evaluation of varying thinning times.

4.4. Residual stress

Different wafer thinning techniques (LT, MT, ECMT, and CMT) lead to significant variations in residual stress. LT technology generates the highest absolute residual stress, with both substantial compressive stress (approximately -650 MPa [18]) and tensile stress (approximately 448.97 MPa [19]). While compressive stress typically improves surface strength and inhibits crack propagation, tensile stress can increase the risk of crack extension. MT technology results in slightly lower residual stress compared to LT, with compressive stress of -340 MPa [20] and tensile stress of 155 MPa. [21] On the other hand, ECMT technology produces lower residual stress, all of which is tensile, at approximately 114.75 MPa [22] and 76.5 MPa [23]. While tensile stress may enhance material plasticity, excessive stress can contribute to crack propagation. CMT technology generates the lowest residual stress, with compressive stress of approximately -101.4 MPa [24] and tensile stress of 70 MPa [25], suggesting that this technique effectively balances stress distribution, minimizing the adverse effects of extreme stress values on the wafer. The differences in the types and magnitudes of residual stress generated by these thinning techniques have a significant impact on the wafer's performance and subsequent processing (see Fig. 9).

Optimising the laser spot shape and energy distribution using DOE technology successfully reduced the residual stress generated during LT processing to levels comparable to those observed with MT. The resulting residual stress is in close alignment with that seen in ECMT, demonstrating the effectiveness of this approach. Further investigation explores how different single-pulse energies and processing cycles impact the residual stress on the wafer surface when using DOE and GAU spots. The results indicate that the residual stress on the surface processed with DOE spots is approximately half of that observed with GAU spots. Although residual stress increases for both spot types as single-pulse energy rises, their behaviours during processing cycles differ. As the number of processing cycles increases, residual stress for both spot types gradually stabilizes. However, the surface residual stress for the DOE spot reaches a stable value earlier, around 50 cycles, while the stress for the GAU spot increases to a maximum before slowly decreasing.

These phenomena can be attributed to several factors. Firstly, DOE spots offer a more uniform energy distribution, which leads to more consistent thermal effects on the wafer surface and results in more stable residual stress that reaches equilibrium more quickly. In contrast, GAU spots exhibit stronger localized energy concentration, creating larger

thermal gradients and more intense thermal stresses during the early stages of processing, leading to a longer adjustment period for the residual stress. Additionally, with more processing cycles, the localized high-temperature effects from the GAU spot cause the surface residual stress to decrease, likely due to the material's thermal diffusion properties and stress relaxation processes. In conclusion, the uniformity of the DOE spot and its lower localized thermal gradients enable it to stabilize the residual stress on the wafer surface more rapidly, whereas the GAU spot requires more cycles to achieve a stable state.

4.5. Surface wettability

The hydrophilicity of the thinned surface plays a crucial role in the subsequent processing of fabricated silicon wafers. For instance, in Backside Wet Etching, a hydrophilic surface enhances the interaction between the etching solution and the wafer surface, leading to improved etching efficiency [26]. To evaluate the wettability of the surface, the water droplet contact angle is measured after thinning with both the DOE-based and GAU laser beams. A contact angle greater than 90° indicates a hydrophobic surface, while a contact angle less than 90° signifies hydrophilicity.

The surfaces thinned with the DOE-based beam exhibited isotropic hydrophilicity, showing similar wettability in both the radial and tangential directions. In contrast, the surfaces thinned with the GAU beam displayed anisotropic hydrophilicity. As the energy increased, the contact angle of the DOE-based beam's thinned surface decreased from 93.04° to 77.82° in the tangential direction and from 86.84° to 75.71° in the radial direction. However, the GAU laser beam-thinned surface showed different behaviour in the two directions. In the radial direction, the hydrophilicity increased, and the contact angle decreased from 96.21° to 88.68° with increasing energy. Conversely, in the tangential direction, the surface became more hydrophobic, with the contact angle increasing from 97.51° to 104.51° (see Fig. 10a).

The surfaces thinned with the DOE-based laser beam lack prominent texture, resulting in a consistent morphology throughout the texture and path (see Fig. 10b). As a result, water droplets spread uniformly in all directions. In contrast, the surface thinned with the GAU laser beam displays a distinct striated texture along the tangential feed direction (see Fig. 10c). The texture, with a certain height, prevents water droplets from diffusing in the tangential direction. As the energy of a single laser pulse increases, the height of the striated texture also increases, further hindering the water droplet diffusion.

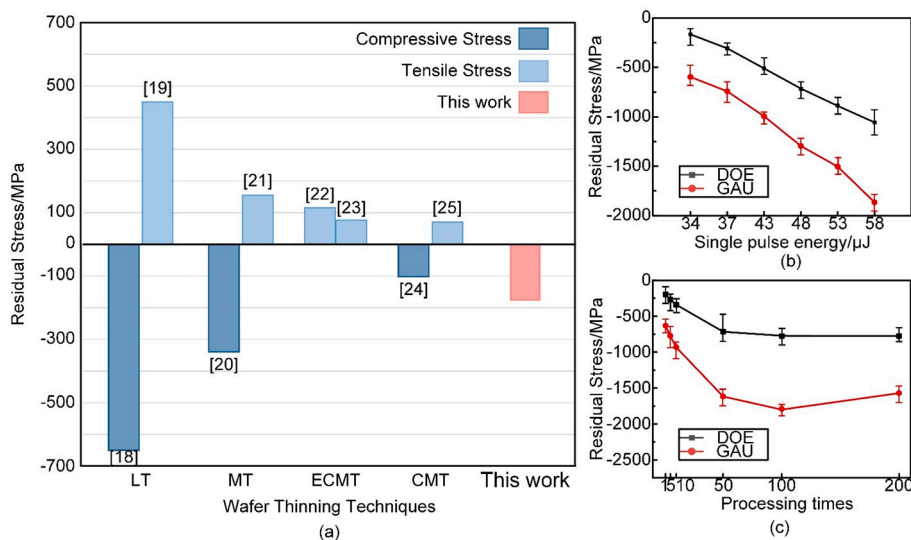


Fig. 9. (a) Compressive stress and tensile stress of different wafer thinning techniques; (b) Residual stress on the surface processed by DOE and GAU beams under different single-pulse energies; (c) Residual stress on the surface processed by DOE and GAU beams after different numbers of processing cycles).

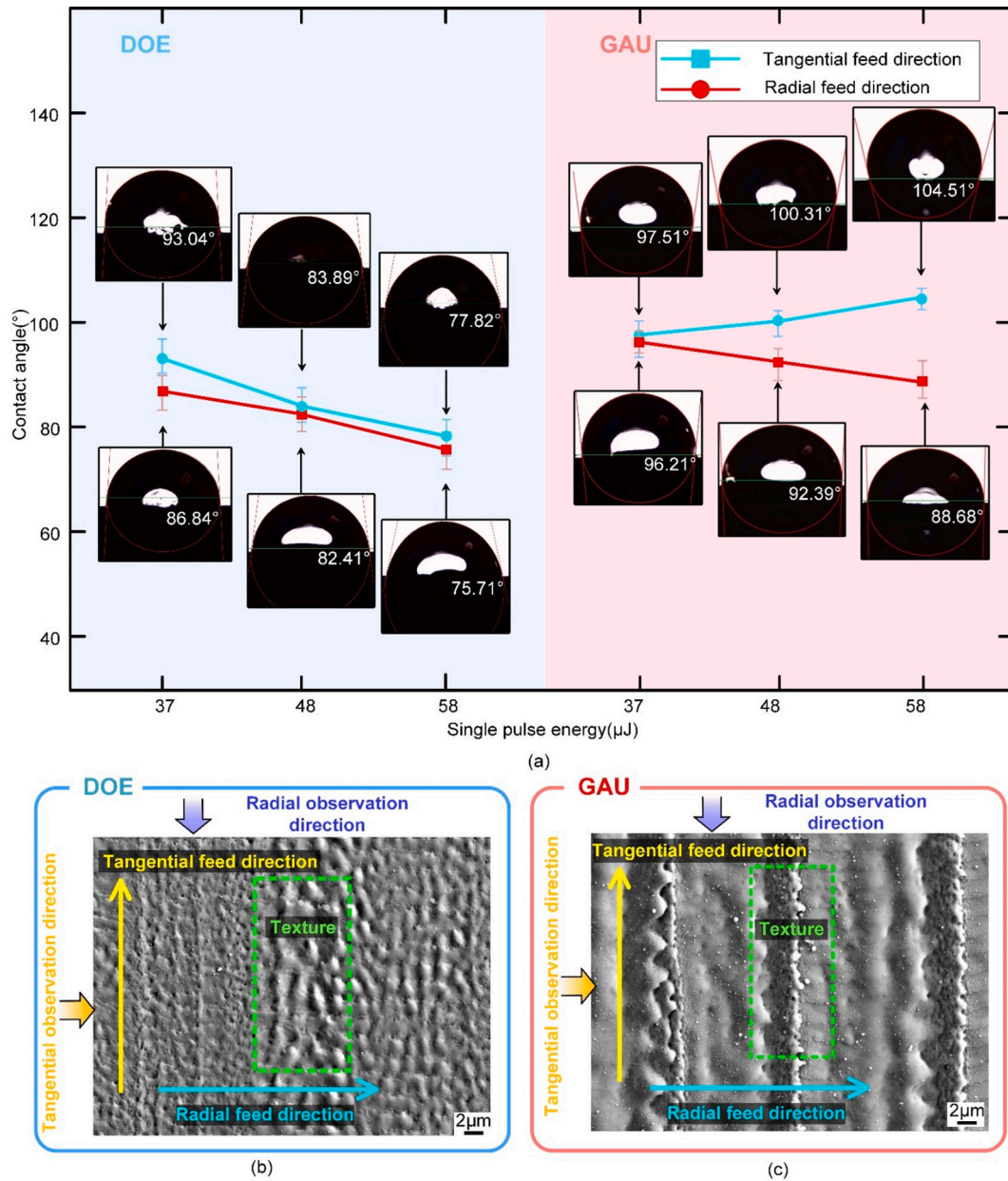


Fig. 10. (a) Contact angles of DOE-based and GAU beam-machined surfaces at different monopoles energies; (b) SEM image of surface texture thinned by GAU beam; (c) EM image displaying the surface texture created by the DOE-based beam.

Surface anisotropy can significantly influence the performance of silicon wafers. For instance, surface roughness shows directional non-uniformity (see Fig. 11a, b). Specifically, the surface is smoother and exhibits lower roughness in the radial direction. In contrast, the axial direction displays a regular peak-valley structure with higher roughness, which could result in increased friction. In the oblique direction, the surface features a wider peak-valley structure, leading to even higher roughness and more complex morphology, which may impact surface performance differently. Additionally, residual stress shows directional non-uniformity (see Fig. 11c). The residual stress in the radial direction is significantly higher than in the axial direction, which could lead to differences in mechanical behaviour across directions, thereby affecting the stability and reliability of the silicon wafer.

The results of this study demonstrate that DOE-based laser thinning offers distinct advantages over the GAU laser method for achieving the desired surface wetting characteristics in silicon wafer thinning. DOE-based laser technology produces isotropic, hydrophilic surfaces,

ensuring uniform contact angles regardless of orientation. These attributes are particularly beneficial for applications like backside wet etching, where consistent wetting ensures homogeneous interaction between the etchant and the silicon surface, leading to consistent etch rates across the entire wafer. This uniformity is crucial for precise dimensional control and minimizing variation in the final product. In contrast, the GAU laser-thinned surfaces exhibit anisotropic hydrophobicity, which is less desirable for uniform wet etching. The pronounced striated texture, although useful for certain specialized applications, hinders uniform wetting and introduces directional dependency.

According to existing studies [27–29], smoother surfaces can potentially be achieved through ultrasonic-assisted thinning, which could help mitigate the anisotropy observed in silicon wafers. This presents a promising approach and perspective for our future investigations, offering a pathway to further optimize the thinning process.

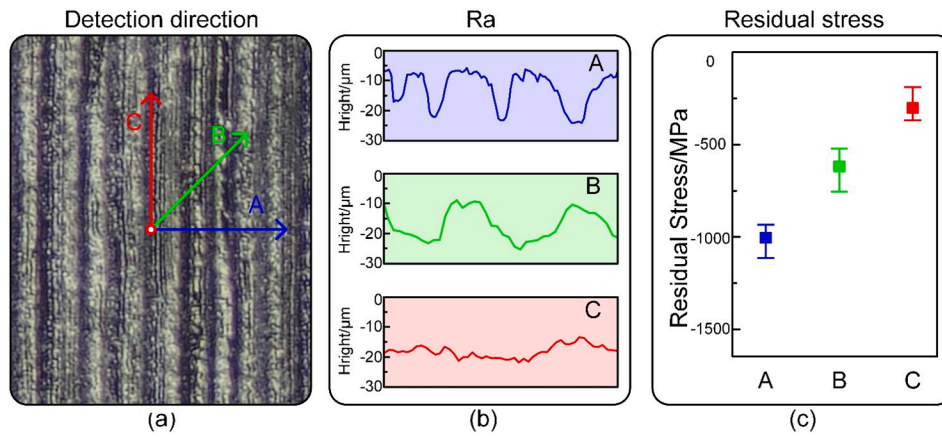


Fig. 11. (a) Different detection directions on the silicon wafer; (b) Surface roughness in different detection directions; (c) Residual stress at different detection directions.

4.6. Different DOE-based ablation mechanisms compared with GAU laser beam

In light of the previously discussed phenomena, DOE-based laser thinning technology shows a distinct advantage in achieving the desired surface wettability for silicon wafer thinning compared to the GAU laser. To further understand this, the ablation mechanisms between the DOE-based laser beam and the GAU laser beam were explored. When thinning on a rotating platform using the DOE-based linear beam, the uniform energy distribution led to reduced thermal effects, enhanced material melting, and minimized vaporization. As a result, solidification of the material occurred, forming micro-spherical ablation pits on the surface (see Fig. 12a). As the platform rotated and the laser scanned laterally, the accumulation of ablation morphology along the laser's feed direction resulted in an irregular textured surface with slight bumps and

undulations (see Fig. 12b). In the radial direction of the laser, the uniform energy distribution of the DOE-based linear spot led to adjacent sections of the ablation tracks exhibiting relatively smooth projections (see Fig. 12c).

In contrast, when thinning was performed using a GAU beam on a rotating platform, the energy was concentrated at the centre of the spot, leading to a sharp, localized rise in temperature and significant thermal effects. This concentrated heat caused material melting or even vaporization, resulting in an ablation pit with a deep centre and shallow edges. Additionally, molten material protruded beyond the edges of the pit (see Fig. 12d). As the platform rotated and the laser scanned laterally (see Fig. 12e), these ablation pits accumulated along the laser's cutting direction, forming a texture reminiscent of fish scales. In the laser's radial direction, adjacent sections of the ablation tracks, located at the edge of the Gaussian pit, displayed shallow ablation depths, resulting in

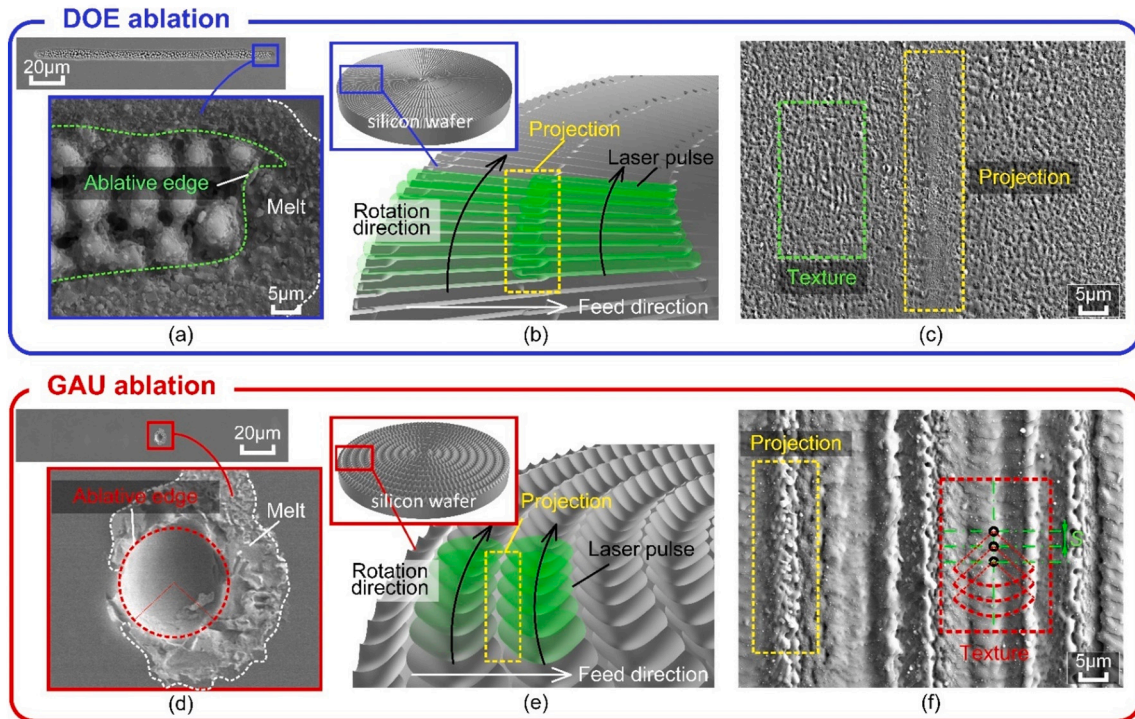


Fig. 12. (a) Single hole ablation morphology of DOE-based beam machining; (b) DOE-based laser thinning diagram; (c) Planar morphology of DOE-based beam ablation; (d) Morphology of single hole ablation in GAU beam machining; (e) Diagram representing GAU laser thinning; (f) Planar morphology obtained from GAU beam ablation.

ridge-like projections (see Fig. 12f).

The observed phenomenon arises from the unique energy distribution properties of DOE-based beams. Unlike GAU beams, which concentrate higher energy at the centre and lower energy at the edges, DOE-based linear spots exhibit a more uniform distribution of energy. This uniform distribution results in a lower energy density at any given point compared to the peak energy density of a GAU beam with equivalent total energy. To quantitatively analyse the relationship between the pulse energy densities of these two beam types, a comparison of the peak energy densities of the two spots is made. The peak energy density of the DOE-based beam can be calculated using Eq. (11) [30].

$$I_D = \frac{\mu E}{lw} \quad (11)$$

where I_D is the DOE-based peak single pulse energy density. μ is the energy transmission coefficient of DOE, which is determined by the transmittance of the material. For sapphire, μ is 98 %. E is the single pulse energy. l is the length of the DOE-based beam at the intersection point, and w is the width of the DOE-based beam at the intersection point. The peak energy density of the GAU can be calculated from Eq. (12)

$$I_G = \frac{2E}{\pi r^2} \quad (12)$$

where I_G is the GAU peak single pulse energy density. r is the radius of the Gaussian beam at the focal spot.

From Table 3, the peak energy density of both beams at the focal spot can be calculated. Comparing the peak energy densities of the two beams at the laser spot, it can be found that the energy ratio of the DOE-based beam to the GAU beam is calculated using Eq. (13)

$$\frac{I_D}{I_G} = \frac{1}{40} \quad (13)$$

Based on this analysis, we can clearly find that the peak single pulse energy density I_D of the DOE-based linear spot is approximately $\frac{1}{40}$ of the Gaussian spot I_G .

5. Non-flat surface thinning with DOE-based laser compared to GAU laser

In addition to flat surfaces, wafer surfaces sometimes exhibit defects (such as scratches and cracks) caused by impacts during transport. Laser thinning is, therefore, necessary to eliminate these defects on uneven surfaces. However, this process presents challenges due to the regenerative effect. When thinning a surface with scratches or cracks, the laser not only reduces the top surface of the wafer but also further thins the bottom of the scratch or crack, thereby widening it in the process. This can lead to the persistence of surface defects even after the desired thinning thickness has been achieved, significantly increasing scrap rates and slowing down processing efficiency. To address this, wafers with ten grooves of gradually increasing depth are fabricated for comparison purposes, using both DOE-based and Gaussian laser beams within the same timeframe (see Fig. 13a).

The DOE-based beam effectively reduced the regenerative effect, with a 41 % reduction in groove depth (from 2.22 μm to 1.31 μm) compared to the GAU beam. In terms of groove width, the DOE-based

beam expanded it from 3.23 μm to 3.62 μm , resulting in a 59 % reduction in the regeneration effect in terms of width compared to the GAU beam (see Fig. 13c). This can be attributed to the avoidance of high peak energy concentrations at the centre and a smaller depth of focus range for the DOE spot, which minimises over-removal of groove edges.

One of the standout advantages of the DOE-based method is its significantly higher processing efficiency. While the GAU beam completes only one thinning pass on the workpiece within a given time, the DOE beam performs ten passes in the same duration (see Fig. 13b). Specifically, the GAU beam reduces the wafer thickness by 3.91 μm in a single pass, while the DOE beam achieves a 5.01 μm thinning in the same period. This represents a remarkable 28.1 % increase in thinning efficiency, demonstrating the DOE beam's superior material removal capability (see Fig. 13c). Consequently, the DOE method delivers ten times the throughput of the GAU method within the same timeframe. The higher number of passes, combined with greater thinning depth, not only accelerates material removal but also ensures a more precise and uniform surface finish, significantly enhancing both thinning efficiency and overall processing accuracy.

This phenomenon can be attributed to the uniform linear distribution of the DOE beam and the optimised spot shape, which allows a more even distribution of laser energy across the workpiece surface during each thinning operation. This uniform distribution avoids the large energy variations between the centre and edges typically observed with conventional GAU beams, resulting in reduced thinning depths and a significant decrease in surface roughness with the DOE beam. The spot length of the DOE beam is ten times longer than that of the GAU beam, leading to a tenfold reduction in thinning time for an equivalent wafer area compared to GAU beams. Therefore, while the GAU beam requires one pass to thin the surface, the DOE beam can achieve this ten times within the same timeframe.

The surface thinned with the DOE-based beam is significantly flatter than that thinned with the GAU beam over the same duration. The DOE-based method achieved a notably low R_z of 198.6 nm, resulting in a smooth and uniform topography (see Fig. 13d). In contrast, the GAU method produced an R_z of 749 nm, indicating a relatively rougher surface with more distinct peaks and valleys (see Fig. 13e).

6. Effects of process parameters in DOE-based laser thinning process

To elucidate the influence of key parameters in the laser thinning process, a series of experiments were designed and conducted, focusing on the effects of (i) single pulse energy and (ii) rotational speed on machining efficiency and surface quality. The experiments utilized single-crystal silicon specimens with dimensions of 10 $\times$ 10 $\times$ 1 mm. By maintaining constant average laser power and radial feed rate, while precisely modulating pulse frequency and rotational speed, the impact of varied thinning conditions on material removal and surface morphology was observed.

- (i) As the single pulse energy (37 to 53 μJ) increased, both the surface roughness and depth also increased. This trend was further confirmed by the reference lines R1 - R4, which represent the maximum depth difference, and D1 - D4, which represent the average machining depth. At a single pulse energy of 53 μJ , the surface roughness reached its peak while achieving the deepest cut (see Fig. 14). This behaviour can be attributed to the fact that, although the spot size remains constant and there is no overlap between adjacent spots, the energy density per pulse increases as the single pulse energy increases. Specifically, when the single pulse energies are 37 μJ , 43 μJ , 48 μJ , and 53 μJ , with a spot area of 200 $\mu\text{m} \times 20 \mu\text{m}$, the corresponding energy densities are 925 J/m², 1075 J/m², 1200 J/m², and 1325 J/m², respectively. The increase in energy density enhances the thermal effects generated by each pulse on the material surface, thereby intensifying the

Table 3
DOE-based and GAU peak energy density calculations.

Laser beam	Length (μm)	Width (μm)	Radius (μm)	Energy transmission coefficient (%)	Peak energy density (J/m ²)
DOE-based	200	20	N/A	98	$2.45 \times 10^8\text{E}$
GAU	N/A	N/A	7.9	N/A	$1 \times 10^{10}\text{E}$

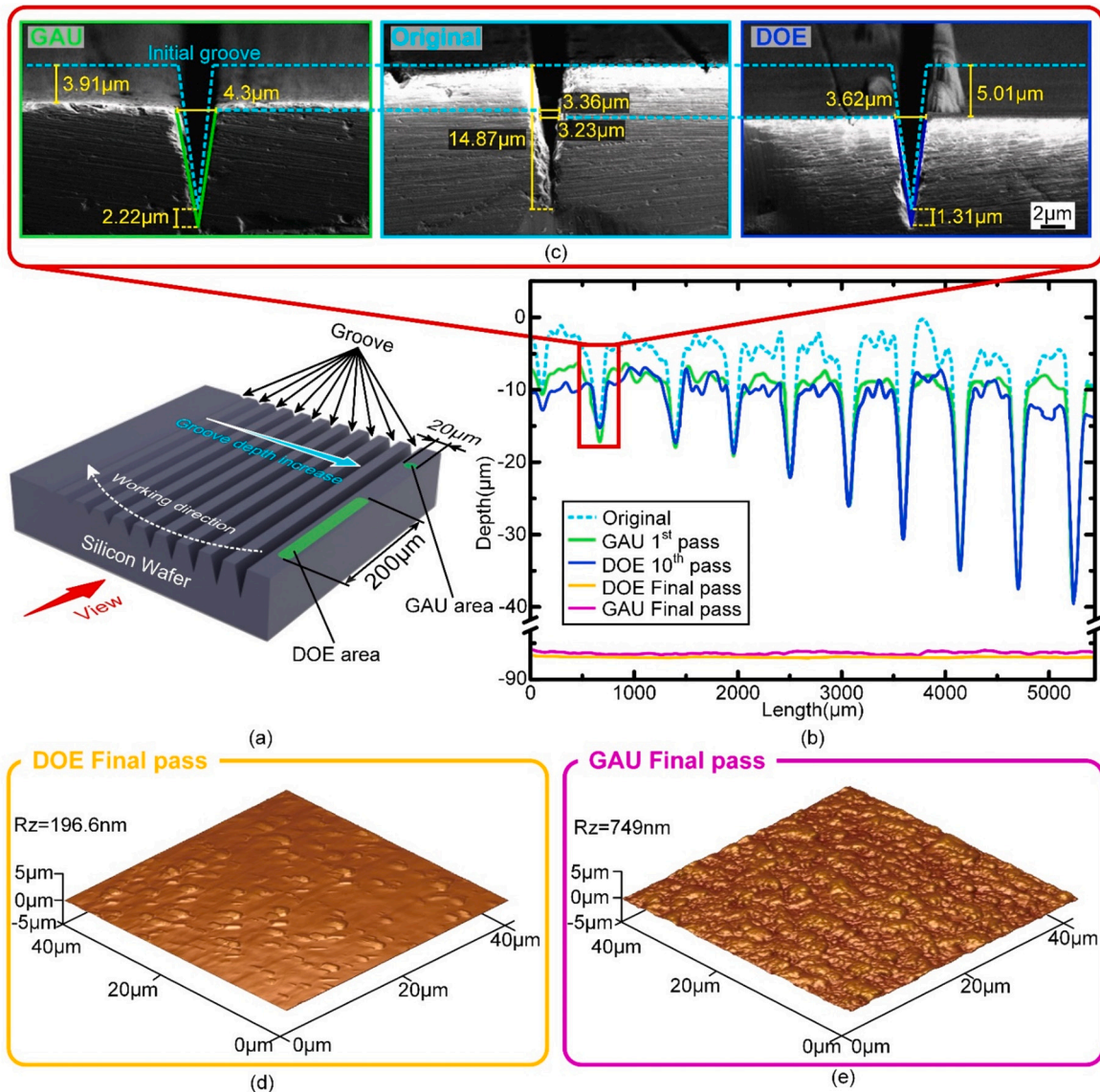


Fig. 13. (a) Groove experiment and laser principle; (b) Depth curve of groove; (c) SEM image showcasing the groove; (d) AFM image capturing the final pass of DOE-based laser thinning; (e) AFM image of GAU laser thinning final pass.

ablation process and improving material removal efficiency. As a result, the amount of material removed per pulse increases, ultimately leading to an increase in cutting depth and surface roughness.

- (ii) As the rotational speed decreased from 229 rpm to 57 rpm, the depth of ablation increased, while the surface roughness decreased. Specifically, lower speeds, such as 57 rpm, resulted in smoother surfaces and greater depths, whereas higher speeds, like 229 rpm, led to irregular surfaces and reduced depths (see Fig. 15). This phenomenon can be attributed to the increase in spot overlap as the rotational speed decreases, which leads to a higher energy density per unit area. When the rotational speed is 229 rpm, the spot overlap is 0, resulting in an energy density of 1075 J/m^2 . As the speed decreases to 172 rpm, 115 rpm, and 57 rpm, the spot overlap increases to 25 %, 50 %, and 75 %, respectively, leading to corresponding increases in energy density to 1343.75 J/m^2 , 1612.5 J/m^2 , and 1881.25 J/m^2 . As the energy density increases, the thermal effects generated by each pulse on the material's surface are enhanced, making the ablation process more uniform and refined. The higher energy density results in smoother material removal, reducing small surface irregularities,

and thus lowering the surface roughness. At the same time, the material removal efficiency improves, leading to an increase in cutting depth while making the surface smoother and reducing roughness.

A comparative analysis was conducted to examine the trends in the average depth of cut and maximum surface roughness (Rz) at different single pulse energies and rotational speeds (see Fig. 16). Both the average depth of cut and Rz increased with rising single pulse energy, indicating that higher energies led to greater material removal and enhanced surface refinement. Conversely, an increase in rotational speed resulted in a decrease in the average depth of cut, while Rz increased. This suggested that lower rotational speeds yield smoother surfaces with larger average depths, whereas higher single pulse energies result in more pronounced surface modifications.

The interaction between single pulse energy and rotational speed is crucial for optimising surface characteristics during laser thinning. Practical applications require a well-balanced selection of these parameters to achieve an ideal compromise between surface roughness and cutting depth, ensuring both high surface quality and efficient thinning. Analysis of the three graphs (see Figs. 14–16) reveals that these

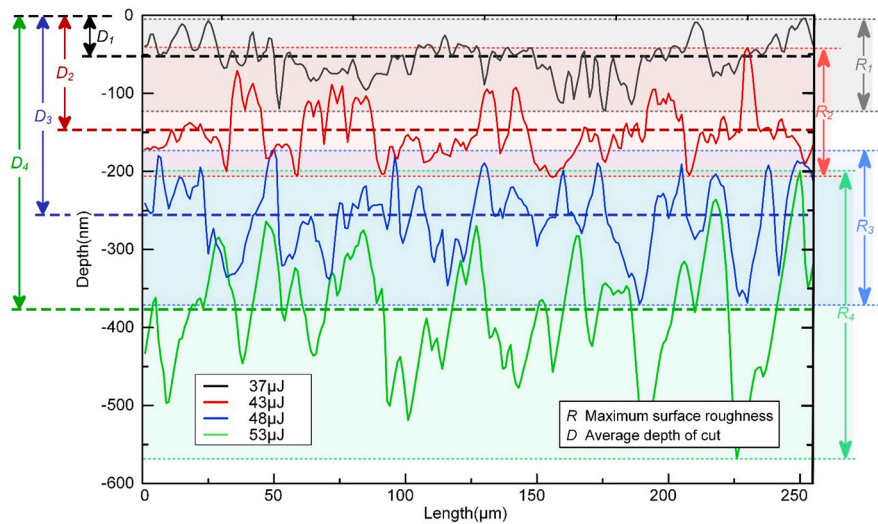


Fig. 14. Surface depth and surface roughness curves of a wafer at different single pulse energy.

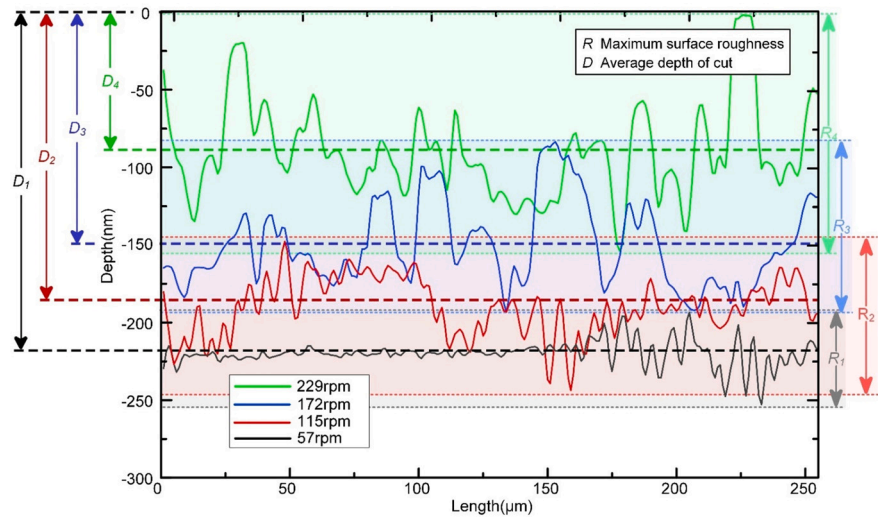


Fig. 15. Surface depth and surface roughness curves of a wafer at rotational speed.

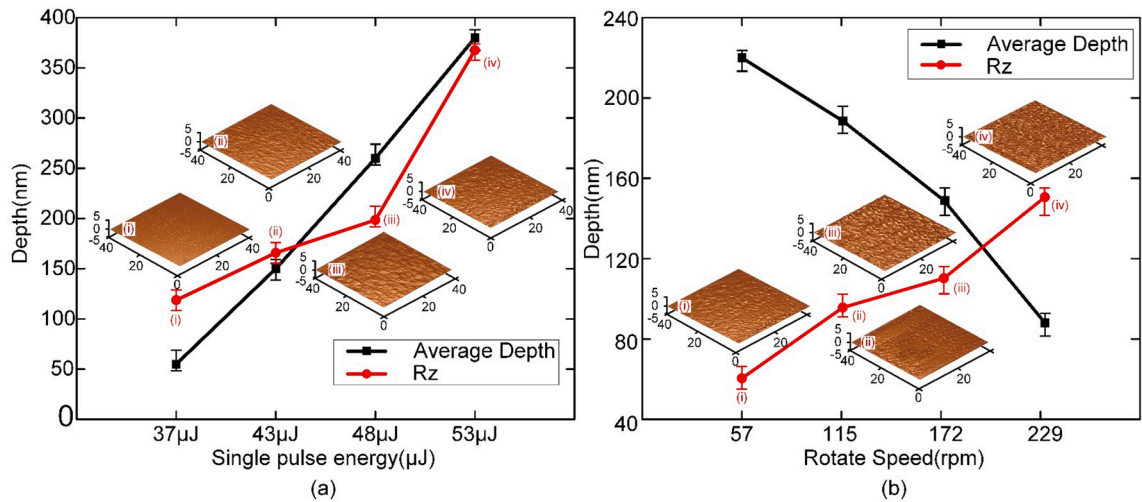


Fig. 16. (a) Changes in mean depth and maximum roughness across varying single pulse energy; (b) Variation of average depth and maximum roughness at different rotational speeds.

parameters have a profound impact on surface quality, highlighting the complexity of laser surface refinement. Effective optimisation is essential for tailoring surface properties to meet specific application needs, whether prioritising smoothness, depth, or a combination of both. These findings underscore the importance of precise control over processing parameters in advanced manufacturing, providing valuable insights that enhance laser manufacturing processes and improve material performance and functionality.

This study compares the proposed laser thinning technology with existing wafer thinning techniques (including MT, CMT, ECMT, and LT) in terms of material removal rate and surface roughness (see Fig. 17).

- (i) In terms of material removal rate, CMT exhibits the lowest rate, ranging from 1 to 12 nm/min. In contrast, ECMT demonstrates a higher material removal rate, ranging from 15 to 75 nm/min. MT further increases the material removal rate to between 40 and 100 nm/min. LT technology achieves the highest material removal rate, reaching between 120 and 500 nm/min.
- (ii) Regarding surface roughness, ECMT offers the lowest surface roughness, ranging from 0.2 to 3 nm, while CMT also produces relatively low surface roughness, ranging from 0.5 to 4 nm. However, MT results in increased surface roughness, with values ranging from 4.4 to 60 nm. LT, on the other hand, exhibits the highest surface roughness, ranging from 45 to 100 nm.
- (iii) The novel laser thinning technology developed in this study successfully achieves a high material removal rate of approximately 450 nm/min by converting the laser beam into a linear spot and employing rotating laser processing. Simultaneously, it maintains a surface roughness of around 45 nm, comparable to that of MT. This technology not only significantly improves wafer processing speed but also ensures relatively low surface roughness, offering an innovative solution for efficient and high-quality wafer production.

7. Conclusion

This research introduces an innovative laser thinning method that employs DOE for the surface thinning of silicon wafers. By converting the laser beam into a linear spot and using rotational laser thinning, this approach effectively overcomes the limitations of conventional GAU laser thinning, which often results in surface defects and non-uniform material removal.

Experimental findings demonstrate the substantial advantages of the DOE-based approach:

1. DOE-based thinning consistently produces smoother surfaces with significantly reduced roughness compared to GAU laser thinning. This improvement is attributed to the uniform energy distribution of the linear spot, which minimises the formation of pits and surface irregularities.
2. DOE-based thinning effectively reduces the formation of silicon dioxide, leading to a more controlled oxidation process. Additionally, it facilitates the creation of a purer single-crystal silicon surface.
3. DOE-based thinning results in more hydrophilic surfaces with uniform textures, offering greater control over surface wettability for specific applications.
4. The larger area coverage of the linear spot significantly enhances processing efficiency compared to the conventional Gaussian spot.
5. DOE-based thinning effectively mitigates the regenerative effect when addressing deeper grooves on the silicon wafer surface, leading to smoother surfaces and reduced material loss.
6. The investigation highlights that pulse frequency and rotational speed significantly impact material removal and surface roughness. Optimising these parameters is crucial for achieving the desired balance between efficiency and surface quality.

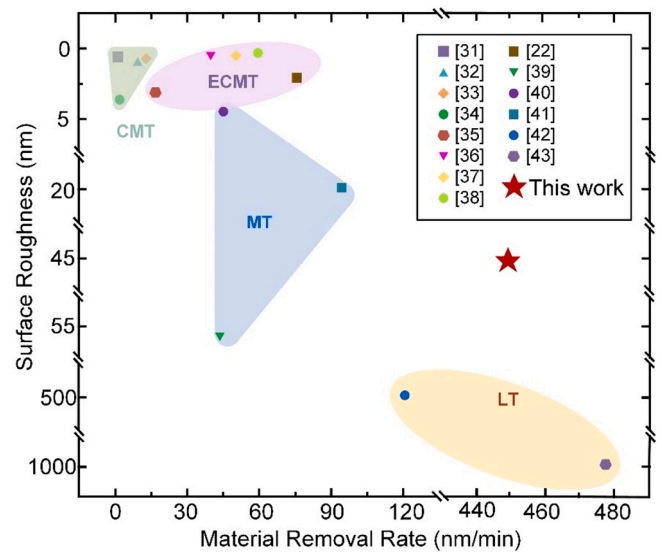


Fig. 17. The MRR and surface roughness of representative wafer thinning [22,31–43].

Overall, the DOE-based rotary laser thinning method presents a promising alternative for achieving high-quality, efficient, and controllable surface precision machining of silicon wafers. This innovative approach has the potential to revolutionise semiconductor device fabrication and other high-precision fields where smooth and precise surfaces are crucial. Future research will focus on further refining the DOE design, optimising processing parameters, and exploring its applicability to other materials and complex geometries.

CRedit authorship contribution statement

Qingfeng Li: Validation, Methodology, Investigation, Conceptualization, Writing – original draft. **Yizheng Wang:** Validation, Writing – original draft. **Haohua Xiu:** Supervision, Writing – review & editing. **Kunyang Li:** Methodology, Investigation. **Chenglong Hua:** Investigation, Formal analysis. **Yongjie Zhao:** Resources, Investigation. **Gongyu Liu:** Supervision, Project administration, Methodology, Writing – review & editing. **Hao Nan Li:** Supervision, Project administration, Conceptualization, Writing – review & editing. **Bixuan Wang:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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